

Selection Sort:

Selection Sort is a simple sorting algorithm that repeatedly finds the smallest element from the unsorted part of the array and places it at the beginning.

Algorithm:

SelectionSort(arr)

1. Start
2. For $i = 0$ to $n - 2$
 - a. Set $min = i$
 - b. For $j = i + 1$ to $n - 1$
If $arr[j] < arr[min]$
 $min = j$
 - c. Swap $arr[i]$ and $arr[min]$
3. Stop

Output:

```
Enter the number of elements: 4
Enter the elements:
2
5
8
6

Sorted Array:
[2, 5, 6, 8]

Execution Time: 0.0000226780 seconds

Time Complexity:
Best Case      :  $O(n^2)$ 
Average Case   :  $O(n^2)$ 
Worst Case     :  $O(n^2)$ 
Space Complexity:  $O(1)$ 

...Program finished with exit code 0
Press ENTER to exit console.
```

Time Complexity:

- **Best Case: $O(n^2)$**
Selection Sort always scans the remaining unsorted elements, even if the array is already sorted.
- **Average Case: $O(n^2)$**
Performs the same number of comparisons regardless of the input order.
- **Worst Case: $O(n^2)$**
The maximum number of comparisons is made for every pass.

Space Complexity:

- **Iterative Selection Sort: $O(1)$**

Uses only a few extra variables (min and a temporary variable for swapping).